Vellore Institute of Technology
CHENNAI

Continuous Assessment Test (CAT) – II - March 2026

Programme	:	B.Tech (CSE & its Specialised Branches)	Semester	:	WS-2025-26
Course Code & Course Title	:	BSCE305L & Embedded Systems	Class Number	:	CH2025260502883 CH2025260502886 CH2025260502888 CH2025260503111 CH2025260503112
Faculty	:	DR. MOHAMMED AARIF K O DR. SUHASINI S DR. LATHA P DR. DHANUSH R DR. GUGAPRIYA G	Slot	:	A2+TA2
Duration	:	90 minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		A telecommunication company implements a lossless data compression technique to efficiently transmit messages over the network. Develop an optimal encoding scheme using a tree-based leaf node approach for the given message: Message: EFFFBBBGGGDBGFFFDDBEEGGBFFF i) Construct the tree using leaf node frequency analysis and generate the corresponding prefix codes.**(5 Marks)** ii) Apply the Run-Length Encoding technique to encode the given message. **(3 Marks)** iii) Compare both methods and determine which technique achieves better compression for this message. Justify your answer. **(2 Marks)**	10	2	K3
2		Develop an Online Food Ordering System that allows customers to browse restaurants, place orders, and make payments. The system should include the following functionalities: User Management: Customers can register, log in, and update their profile details such as address and contact number. Restaurant Management: The system maintains information about restaurants including restaurant name, location, and menu items. Menu Management: Each restaurant can add, update, or remove food items from their menu. Each item contains details such as item name, price, and availability. Order Processing: Customers can place orders, track order status, and cancel orders before preparation begins. Payment Handling: The system processes payments through different methods such as credit card, debit card, or digital wallets.	10	2	K3

<table>
<tr><td></td><td></td><td>Admin Controls: Administrators can manage restaurants, monitor orders, and handle user accounts.
Identify the essential system components (classes) required for this system.
Design an UML Class Diagram showing the classes, attributes, methods, and relationships among them.</td><td></td><td></td><td></td></tr>
<tr><td>3</td><td></td><td>A Hospital Appointment Management System that allows patients to book appointments with doctors.
System functionalities include:
• Patients can create accounts and log in.
• Doctors maintain schedules and specialization details.
• Patients can schedule, reschedule, or cancel appointments.
• Doctors can add diagnosis notes and prescriptions.
• Administrators manage doctor profiles and appointments.
Identify the actors involved in the system & major use cases.
Draw an UML representing the interactions between actors and the system.(5 Marks)
Draw an Activity diagram for the Hospital Appointment Management System.(5 Marks)</td><td>10</td><td>2</td><td>K2</td></tr>
<tr><td>4</td><td></td><td>An automotive ECU performs the following tasks which is given in Table.1
Table 1
<table><tr><th>Task</th><th>Description</th><th>Computation Time (C)</th><th>Deadline</th></tr><tr><td>ENG</td><td>Engine Monitoring</td><td>1</td><td>4</td></tr><tr><td>BRA</td><td>Brake Control</td><td>2</td><td>8</td></tr><tr><td>DIAG</td><td>Diagnostics</td><td>4</td><td>16</td></tr></table>i) Discuss the priority assignment of the tasks. (1Mark)
ii) Perform RMS schedulability test. (3 Marks)
iii) If utilization bound fails, verify using response time analysis. (5 Marks)
iv) State whether the system is fully schedulable. (2 Marks)</td><td>10</td><td>4</td><td>K3</td></tr>
<tr><td>5</td><td></td><td>Assume that the following tasks are real time periodic tasks shown in Table.2
Table.2
<table><tr><th>Task</th><th>Computation Time (C)</th><th>Deadline</th><th>Period (T)</th></tr><tr><td>T1</td><td>2</td><td>6</td><td>8</td></tr><tr><td>T2</td><td>1</td><td>4</td><td>5</td></tr><tr><td>T3</td><td>3</td><td>8</td><td>10</td></tr></table>i) Calculate the total CPU utilization of the given task set. (3 Marks)
ii) Determine whether the task set is schedulable using the Earliest Deadline First (EDF) scheduling algorithm. (3 Marks)
iii) Draw the Gantt chart for one hyper period to justify your answer. (4 Marks)</td><td>10</td><td>4</td><td>K3</td></tr>
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********All the best **********